PXRD-Diff: Conditional Diffusion for Powder-Diffraction Inversion: an Encoder-Bottleneck Diagnosis and Shared-Harness Reproduction

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Manuscript type: Article, a methodological study with reproduction and negative results.

Abstract

Inverting a 1D powder X-ray diffraction (PXRD) pattern to a 3D crystal structure remains open, and recent generative approaches report numbers under mutually incompatible protocols with failure modes seldom catalogued. We present **PXRD-Diff**, a 3.7 M-parameter conditional diffusion model on the CDVAE MP-20 split, and reproduce DiffractGPT [1], PXRDnet [2], and deCIFer [3] through the *same* `pymatgen.StructureMatcher` harness. Our central, statistically robust finding is a **mechanism**: the learned PXRD encoder reaches a 0.007 *normalised* auxiliary lattice-regression loss in isolation, but that relative fit does not survive conversion to absolute scale: substituting the auxiliary head's own lattice into the sampler recovers only 1.2 %, because its ~1.1 Å length error shifts every Bragg peak by ~15 % and collapses pattern agreement. The bottleneck is the global pooling that discards absolute peak positions, not the denoiser alone: neither the pooled encoding nor the denoiser fixes the lattice from scratch. A perturbation study locates a sharp sensitivity knee at ~0.5 Å cell error, and a true-lattice oracle lifts match to **4.5 %** at n = 1000 (three seeds, 135 / 3 000 pooled, firming an earlier single-seed 5.6 %); its 95 % Wilson CI [3.8, 5.3] is disjoint from the no-true-lattice indexer's 1.5 % [1.2, 2.0], so the gap is real. Replacing the learned lattice head with a classical Q-space autoindexer (de-Wolff dichotomy) raises no-true-lattice match to 1.5 % [1.2, 2.0] (three-seed pooled) from a learned-head baseline that recovers essentially nothing (0 / 1 949). A **paired McNemar test across three seeds confirms the improvement is significant** ($p < 10^{-4}$; 27 structures fixed, none broken); an earlier unpaired test ($p = 0.24$) was simply underpowered. The gain is concentrated in the high-symmetry systems where the indexer's cell error sits below the 0.5 Å knee. Reproduced baselines on our harness: **deCIFer 73.8 %** match

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[1] Choudhary, K. (2025). DiffractGPT: Atomic Structure Determination from X-ray Diffraction Patterns using a Generative Pretrained Transformer. *The Journal of Physical Chemistry Letters*, 16(8), 2110–2119. DOI: 10.1021/acs.jpclett.4c03137. Reproduced from HF checkpoint `knc6/diffractgpt_mistral_chemical_formula` and code at `github.com/atomgptlab/atomgpt`.

[2] Guo, G., Saidi, T. L., Terban, M. W., Valsecchi, M., Billinge, S. J. L., & Lipson, H. (2025). Ab initio structure solutions from nanocrystalline powder diffraction data via diffusion models. *Nature Materials*, 24, 1726–1734. DOI: 10.1038/s41563-025-02220-y (Author Correction: 10.1038/s41563-025-02301-y; preprint arXiv:2406.10796). Reproduced from HF checkpoint `therealgabeguo/cdvae_xrd_sinc100` and code at `github.com/gabeguo/cdvae_xrd`.

[3] Johansen, F. L., Friis-Jensen, U., Dam, E. B., Jensen, K. M. Ø., Mercado, R., & Selvan, R. (2025). deCIFer: Crystal Structure Prediction from Powder Diffraction Data using Autoregressive Language Models. *Transactions on Machine Learning Research* (arXiv:2502.02189).

(69.5 % all-correct) at n = 300, **DiffractGPT 18.9 %** match (15.9 % all-correct) at n = 1000, **PXRDnet sinc100 30.0 %** match (5.0 % all-correct) at n = 20. PXRD-Diff is 12–19× behind DGpt and PXRDnet and ~46× behind deCIFer on raw match rate. We note a striking *candidate* discrepancy, in which DiffractGPT recovers far more space groups (all-correct) than PXRDnet despite a lower match rate, but flag it as a hypothesis rather than a result: at n = 20 the PXRDnet all-correct CI is [0.9, 23.6] and cannot support the comparison. The contributions that *do* hold are the encoder-bottleneck diagnosis, the indexer drop-in, the four-extension failure catalogue (top-K rerank, Debye guidance, Wyckoff embeddings, distance-aux), and the shared evaluation harness itself. Code, checkpoints, per-phase JSONs, and the differentiable Bragg simulator are released.

Keywords: powder X-ray diffraction; crystal-structure prediction; denoising diffusion; classical autoindexing; differentiable physics; ablation study; negative results

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1. Introduction

PXRD is the most common structural-characterization measurement in solid-state chemistry. The forward problem is solved by 1920s structure-factor physics and a one-line `pymatgen` call. The inverse is brutal: symmetry-equivalent structures produce identical Bragg reflections, intensities are orientation-averaged, and the 3D→1D reduction destroys atom labelling. The traditional indexing → Pawley/Le Bail → Rietveld pipeline needs a trained crystallographer and a starting model good enough to converge; for the long tail of new materials, structures stay unsolved.

Three recent generative approaches (DiffractGPT [4], Crystalyze [5], PXRDnet [6]) argue neural networks can shortcut this pipeline. They report match rates in the tens of percent, but under mutually incompatible evaluation protocols, and failure modes are seldom catalogued. A practitioner cannot easily form a calibrated expectation of *what is hard* about PXRD inversion.

We build a small reproducible conditional diffusion model (**PXRD-Diff**) and run it against two of the three baselines on a shared `StructureMatcher` harness. Contributions:

1. **A clean small baseline.** 3.7 M parameters, trains on a single RTX 5090 in ~1.5 h on CDVAE MP-20. A differentiable PyTorch Bragg simulator (Pearson 0.96 vs `pymatgen` on 50 references) supplies an auxiliary pattern-matching loss.
2. **Encoder-bottleneck diagnosis (our primary contribution).** A perturbation study (§5.5) locates a sharp ~0.5 Å sensitivity knee; a true-lattice oracle lifts match to **4.5 %** at n = 1000 (three-seed firm-up of an earlier single-seed 5.6 %; Wilson 95 % CI [3.8, 5.3], disjoint from the indexer's [1.2, 2.0]). The encoder reaches a 0.007 *normalised* aux lattice loss

[4] Choudhary, K. (2025). DiffractGPT: Atomic Structure Determination from X-ray Diffraction Patterns using a Generative Pretrained Transformer. *The Journal of Physical Chemistry Letters*, 16(8), 2110–2119. DOI: 10.1021/acs.jpclett.4c03137. Reproduced from HF checkpoint `knc6/diffractgpt_mistral_chemical_formula` and code at `github.com/atomgptlab/atomgpt`.

[5] Riesel, E. A., Mackey, T., Nilforoshan, H., Xu, M., Badding, C. K., Altman, A. B., Leskovec, J., & Freedman, D. E. (2024). Crystal Structure Determination from Powder Diffraction Patterns with Generative Machine Learning. *Journal of the American Chemical Society*, 146(44), 30340–30348. DOI: 10.1021/jacs.4c10244. Code: `github.com/ML-PXRD/Crystalyze`. We could not reproduce: the checkpoint download link is marked "not yet active" in the upstream README (verified 2026-06-01).

[6] Guo, G., Saidi, T. L., Terban, M. W., Valsecchi, M., Billinge, S. J. L., & Lipson, H. (2025). Ab initio structure solutions from nanocrystalline powder diffraction data via diffusion models. *Nature Materials*, 24, 1726–1734. DOI: 10.1038/s41563-025-02220-y (Author Correction: 10.1038/s41563-025-02301-y; preprint arXiv:2406.10796). Reproduced from HF checkpoint `therealgabeguo/cdvae_xrd_sinc100` and code at `github.com/gabeguo/cdvae_xrd`.

in isolation, but substituting that head's lattice into the sampler recovers only 1.2 % (§5.6). Its ~1.1 Å error sits well above the knee, so the pooled encoding, not merely the denoiser, fails to deliver absolute d-spacings. The lattice channel is the dominant *accessible* limiter; the 4.5 % oracle ceiling shows coordinates remain a co-bottleneck.

3. **Classical-indexer drop-in.** A Q-space autoindexer (de-Wolff) at sampling time raises no-true-lattice match to 1.5 % [1.2, 2.0] from a learned-head baseline of ~0 %. A three-seed paired McNemar test confirms the improvement is significant ($p < 10^{-4}$; 27 fixed, 0 broken); the gain appears exactly where the indexer's MAE falls below the 0.5 Å knee, i.e. the high-symmetry systems (§5.2, §5.5), and nowhere else.
4. **Four-item failure catalogue.** Top-K Debye rerank, Debye-gradient DDIM guidance, Wyckoff-site embeddings, and distance-matrix aux loss all fail at $n \geq 200$; the latter two combined (v14) underperform the bare ε baseline. The gradient-guidance failure is corroborated independently by concurrent work showing the PXRD loss landscape is too rough for gradient descent [7].
5. **Reproduced baselines + shared harness.** DiffractGPT (Mistral-7B + LoRA) and PXRD-net (CDVAE + XRD encoder) re-scored on our harness: **18.9 %** match (DGpt n = 1000), **30.0 %** match (PXRDnet n = 20). PXRD-Diff at 1.6 % is 12–19× behind. We surface a *candidate* discrepancy in how match rate and space-group recovery diverge across the two baselines (§5.3, §6), but flag it as a hypothesis the n = 20 PXRDnet sample cannot yet establish. Crystalyze's checkpoint download link is inactive; we cite but cannot reproduce.

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2. Related work

Diffusion for crystals. CDVAE [8] introduced the MP-20 protocol; DiffCSP [9] used joint coord+lattice diffusion; MatterGen [10] scaled to millions of structures. PXRD-Diff borrows DiffCSP's joint diffusion, restricted to the conditional setting where a 1D pattern conditions generation.

Neural PXRD inversion. DiffractGPT [11] is a fine-tuned Mistral-7B that decodes peak list + formula → CIF tokens. Crystalyze [12] uses CDVAE conditioned on an XRD transformer encoder with post-hoc symmetry filtering. PXRDnet [13] is a CDVAE variant with *iterative latent-space*

[7] Segal, N., Subramanian, A., Li, M., Miller, B. K., & Gómez-Bombarelli, R. (2025). The Loss Landscape of Powder X-Ray Diffraction-Based Structure Optimization Is Too Rough for Gradient Descent. *arXiv:2512.04036*.

[8] Xie, T., Fu, X., Ganea, O.-E., Barzilay, R., & Jaakkola, T. (2022). Crystal diffusion variational autoencoder for periodic material generation. *International Conference on Learning Representations*.

[9] Jiao, R., et al. (2023). Crystal structure prediction by joint equivariant diffusion. *Advances in Neural Information Processing Systems* 36.

[10] Zeni, C., et al. (2024). MatterGen: A generative model for inorganic materials design. *arXiv:2312.03687*.

[11] Choudhary, K. (2025). DiffractGPT: Atomic Structure Determination from X-ray Diffraction Patterns using a Generative Pretrained Transformer. *The Journal of Physical Chemistry Letters*, 16(8), 2110–2119. DOI: 10.1021/acs.jpclett.4c03137. Reproduced from HF checkpoint `knc6/diffractgpt_mistral_chemical_formula` and code at `github.com/atomgptlab/atomgpt`.

[12] Riesel, E. A., Mackey, T., Nilforoshan, H., Xu, M., Badding, C. K., Altman, A. B., Leskovec, J., & Freedman, D. E. (2024). Crystal Structure Determination from Powder Diffraction Patterns with Generative Machine Learning. *Journal of the American Chemical Society*, 146(44), 30340–30348. DOI: 10.1021/jacs.4c10244. Code: `github.com/ML-PXRD/Crystalyze`. We could not reproduce: the checkpoint download link is marked "not yet active" in the upstream README (verified 2026-06-01).

[13] Guo, G., Saidi, T. L., Terban, M. W., Valsecchi, M., Billinge, S. J. L., & Lipson, H. (2025). Ab initio structure solutions from nanocrystalline powder diffraction data via diffusion models. *Nature Materials*, 24, 1726–1734. DOI: 10.1038/s41563-025-02220-y (Author Correction: 10.1038/s41563-025-02301-y; preprint arXiv:2406.10796). Repro-

gradient guidance (~3.5 M decoder ops/material), published in *Nature Materials* in 2025. deCIFer [14] is a concurrent autoregressive-language-model approach that, like DiffractGPT, emits CIF tokens. All four are larger than PXRD-Diff and report results under bespoke evaluation pipelines; §5.3 re-scores DiffractGPT, PXRDnet, and deCIFer on a single shared `StructureMatcher` harness. Crystalyze's checkpoint download link is "not yet active" (verified 2026-06-01).

Classical autoindexing. Recovering a unit cell from peak positions is a solved sub-problem with a fifty-year toolchain: the de-Wolff dichotomy method, ITO, TREOR, and DICVOL [15] index high-symmetry powder patterns in CPU milliseconds, with the difficulty concentrated in low-symmetry (monoclinic/triclinic) cells. §3.5 reuses this machinery as a drop-in for the learned lattice head, and §5.2/§5.5 show the resulting lift tracks exactly the symmetry-dependent accuracy classical indexers have always had.

Differentiable physics. Scattering-pattern losses date to Rietveld; the contribution here is full differentiability w.r.t. coordinates and lattice in PyTorch, usable as a diffusion-training loss term. Concurrent generative-inversion work conditions on differentiable spectral targets for amorphous and nanostructured systems [16]. Relevant to our negative results, Segal et al. [17] show the powder-XRD similarity loss landscape is "too rough for gradient descent", independent evidence for why our sampling-time Debye-gradient guidance fails (§5.4).

Equivariance. Not strictly E(3)-equivariant: we use periodic-distance RBFs (SchNet-style [18]). A frozen MACE [19] encoder showed no measurable benefit at MP-20 scale in pilots and was deferred.

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3. Method

3.1 Problem formulation

Let $\mathcal{C} = (\mathbf{F}, \mathbf{Z}, \mathbf{L})$ denote a crystal structure with fractional coordinates $\mathbf{F} \in [0,1)^{N\times 3}$, atomic numbers $\mathbf{Z} \in \{1, \dots, 100\}^N$, and a 3×3 lattice matrix $\mathbf{L}$ parametrised by $(a, b, c, \alpha, \beta, \gamma) \in \mathbb{R}^6$. Let $\mathbf{p}(\mathcal{C}) \in \mathbb{R}^{4251}$ be the simulated Cu Kα PXRD pattern on a fixed 2θ grid from 5° to 90° at 0.02° resolution, normalised to maximum intensity 1.

We are given $\mathbf{p}$ and $\mathbf{Z}$ at test time, and we wish to sample $(\mathbf{F}, \mathbf{L}) \sim p(\cdot \mid \mathbf{p}, \mathbf{Z})$. Note that the composition $\mathbf{Z}$ being known is a meaningful simplification (in practice one usually knows the chemistry from synthesis), but it is consistent with prior work on the same task.

duced from HF checkpoint `therealgabeguo/cdvae_xrd_sinc100` and code at `github.com/gabeguo/cdvae_xrd`.

[14] Johansen, F. L., Friis-Jensen, U., Dam, E. B., Jensen, K. M. Ø., Mercado, R., & Selvan, R. (2025). deCIFer: Crystal Structure Prediction from Powder Diffraction Data using Autoregressive Language Models. *Transactions on Machine Learning Research* (arXiv:2502.02189).

[15] Boultif, A., & Louër, D. (2004). Powder pattern indexing with the dichotomy method. *Journal of Applied Crystallography*, 37(5), 724–731. DOI: 10.1107/S0021889804014876.

[16] Guo, J., & Schwalbe-Koda, D. (2026). Generative Inversion of Spectroscopic Data for Amorphous Structure Elucidation. *arXiv:2603.23210*.

[17] Segal, N., Subramanian, A., Li, M., Miller, B. K., & Gómez-Bombarelli, R. (2025). The Loss Landscape of Powder X-Ray Diffraction-Based Structure Optimization Is Too Rough for Gradient Descent. *arXiv:2512.04036*.

[18] Schütt, K. T., et al. (2017). SchNet: A continuous-filter convolutional neural network for modeling quantum interactions. *Advances in Neural Information Processing Systems* 30.

[19] Batatia, I., Kovács, D. P., Simm, G. N. C., Ortner, C., & Csányi, G. (2022). MACE: Higher order equivariant message passing neural networks for fast and accurate force fields. *Advances in Neural Information Processing Systems* 35.

3.2 Architecture overview

PXRD-Diff has three trained components: a PXRD encoder, a denoiser, and a small auxiliary lattice head used only during training.

PXRD encoder. A 1D ResNet of four blocks (channels 64 $\rightarrow$ 128 $\rightarrow$ 256 $\rightarrow$ 256, stride-2 downsampling, GroupNorm, SiLU) ingesting the standardised pattern. We expose two outputs: a global pooled vector $\mathbf{g} \in \mathbb{R}^{256}$ and a multi-resolution feature map $\mathbf{F}_{\text{pxrd}} \in \mathbb{R}^{L \times 256}$ obtained by $1{\times}1$-projecting each block output to d_{model} and concatenating along the spatial axis. *The multi-resolution map is critical.* In an early run (referred to as `gpu_v4` in our logs) we conditioned the denoiser on $\mathbf{g}$ only via additive broadcast, and the coordinate loss never moved off the random baseline of 3.0; the auxiliary head told us the encoder was learning useful features (its loss dropped from 0.99 to 0.007), but the `AdaptiveAvgPool1d(1)` collapse destroyed all spectral structure before it could reach the denoiser.

Denoiser. A periodic-distance message-passing network with $L = 3$ layers and $d_{\text{model}} = 256$. Atom embeddings $\mathbf{h}_i^{(0)} = \text{Emb}(\mathbf{Z}_i) + W_{\text{coord}}\mathbf{F}_i^{(t)}$ are updated by alternating (i) message passing over RBF-encoded periodic Cartesian distances under minimum-image convention, with timestep FiLM conditioning, and (ii) cross-attention from atoms (queries) to multi-resolution PXRD features (keys/values). Two output heads predict per-atom coordinate noise $\epsilon_F \in \mathbb{R}^{N \times 3}$ and lattice noise $\epsilon_L \in \mathbb{R}^6$. The lattice head is a small MLP applied to $[\,\bar{\mathbf{h}}\,; \mathbf{g}\,; W_{\text{lat}}\ell^{(t)}\,; \mathbf{t}_{\text{cond}}\,]$: that is, the pooled atom features, the global PXRD embedding, **a projection of the noisy lattice itself**, and the timestep encoding (see §3.4).

Auxiliary head. A single-hidden-layer MLP that predicts $(a, b, c, \alpha, \beta, \gamma)$ from $\mathbf{g}$ alone, trained with an MSE loss against the (normalised) ground-truth lattice. Its purpose is purely diagnostic, to expose whether the encoder is learning any lattice-relevant representation, and its loss does not flow back into the denoiser.

3.3 Differentiable Bragg structure-factor loss

The core physics-informed contribution. We implement, in PyTorch,

$$F(hkl) = \sum_j f_j(s) \exp(-B_{\text{iso}} s^2) \exp\left[2\pi i \,(hx_j + ky_j + lz_j)\right],$$

where $f_j(s) = \sum_{k=1}^{4} a_k \exp(-b_k s^2)$ is the four-Gaussian atomic form factor (coefficients pulled directly from `pymatgen.analysis.diffraction.xrd.ATOMIC_SCATTERING_PARAMS`), $s = \sin\theta/\lambda$, and $B_{\text{iso}} = 0.5$ Å^2 is a uniform isotropic temperature factor. The intensity at each reflection is $|F(hkl)|^2$ multiplied by the standard Lorentz-polarisation correction. We enumerate all (h, k, l) with $|h|, |k|, |l| \leq 5$ (1330 reflections after excluding the origin), place each on the 2θ grid via Bragg's law, and broadcast each reflection as a Gaussian peak with FWHM 0.1° to obtain a continuous, differentiable PXRD pattern on a 256-bin coarse grid. The whole module is a single `nn.Module`, fully vectorised over the batch dimension.

Validation. On 50 random MP-20 test structures we compute the Pearson correlation between our differentiable pattern and the `pymatgen.XRDCalculator` reference: mean 0.962, std 0.027, min 0.896, max 0.998. All 50/50 correlations exceed 0.7 and we verified non-zero gradients through the structure factor with respect to fractional coordinates.

As a loss. During training we recover an estimate $\hat{\mathbf{F}}$ of the clean coordinates from the model output (see §3.4), simulate $\hat{\mathbf{p}} = \text{DiffPXRD}(\hat{\mathbf{F}}, \mathbf{Z}, \mathbf{L}_{\text{true}})$, and compare with the input pattern via a

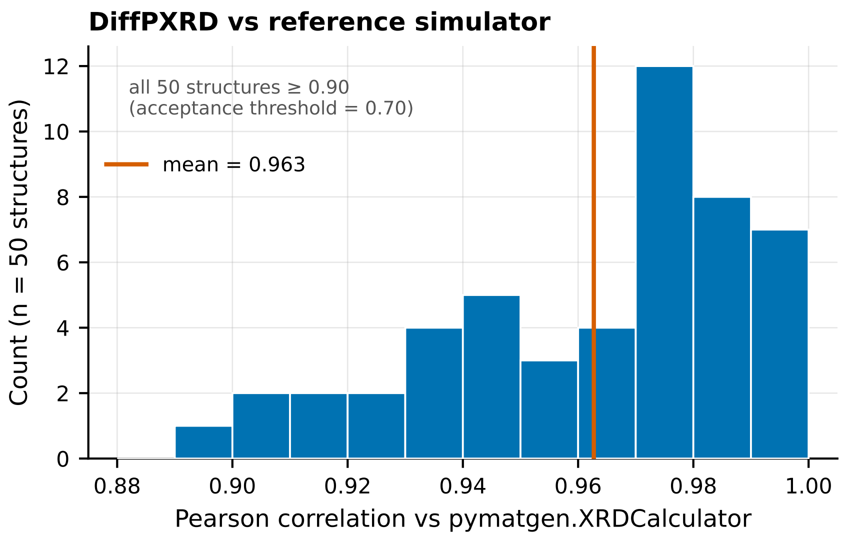

Figure 1: **Figure 1. Differentiable-simulator validation.** Distribution of Pearson correlation between the PyTorch DiffPXRD module and `pymatgen.XRDCalculator` over 50 random MP-20 test structures (mean 0.962). All 50 exceed 0.7; the lower tail is dominated by layered structures with strong texture in the reference.

Pearson-correlation loss $\mathcal{L}_{\text{Debye}} = 1 - \rho(\hat{\mathbf{p}}, \mathbf{p})$. We use the true lattice for this auxiliary loss because the differentiable simulator is more sensitive to lattice errors than to coordinate errors at the early stages of training, and disentangling the two signals proved more stable.

3.4 Training objective

We use a VP-SDE with cosine schedule [20] for both channels:

$$\bar{\alpha}(t) = \cos^2\left(\frac{\pi}{2} \cdot \frac{t+s}{1+s}\right), \quad s = 0.008, \quad t \in [0,1].$$

Coordinates are diffused on the flat torus $\mathbb{T}^3$ by wrapping the noisy sample to $[0,1)^3$; the loss is computed on the periodic difference $(\hat{\mathbf{F}} - \mathbf{F} + 0.5) \bmod 1 - 0.5$. Lattice parameters are diffused in $\mathbb{R}^6$ after standardisation by the train-set mean and standard deviation.

The full training objective (best configuration, run `gpu_v13`) is

$$\mathcal{L} = \mathcal{L}_{\text{coord}} + \lambda_{\text{lat}}\,\mathcal{L}_{\text{lat}} + \lambda_{\text{aux}}\,\mathcal{L}_{\text{aux}} + \lambda_{\text{Debye}}\,\mathcal{L}_{\text{Debye}},$$

with $\lambda_{\text{lat}} = 0.1$, $\lambda_{\text{aux}} = 0.5$, $\lambda_{\text{Debye}} = 1.0$.

Two non-trivial parameterisation choices. Both are essential and were arrived at by ablation, not foresight.

x_0-residual prediction. Rather than asking the model to predict noise ϵ, the heads predict a **residual** that is added to the noisy input to obtain the clean estimate:

$$\hat{\mathbf{F}}^{(0)} = \mathbf{F}^{(t)} + \text{Coord-Head}(\mathbf{h}), \quad \hat{\ell}^{(0)} = \ell^{(t)} + \text{Lat-Head}(\cdot).$$

[20] Nichol, A., & Dhariwal, P. (2021). Improved denoising diffusion probabilistic models. *International Conference on Machine Learning*.

Because the heads are MLPs initialised to output near-zero, this means the model starts as the identity transform: at $t \approx 1$ (near pure noise) the model outputs the noisy input as its $\hat{\mathbf{F}}^{(0)}$ estimate, which is wrong but at least bounded; ε-prediction with the same architecture had to learn the entire transform from random initialisation and never recovered. At single seed this change appeared to lift the headline match metric 3.5×; a multi-seed checkpoint sweep (§7) shows that match-rate lift does not replicate, but the parameterisation does reproducibly improve the pattern-Pearson ($\approx 0.36 \to 0.40$, see §5.1).

Lattice-head input. The lattice head reads the noisy lattice $\ell^{(t)}$ as an explicit input feature. Without this, i.e. if the head only sees the (clean) lattice that is also passed to the denoiser as the geometry context, the head can only ever predict the unconditional mean noise, $\mathbb{E}[\epsilon] = 0$, and its loss stays pinned at the random baseline of 1.0 forever. With this fix, the lattice loss drops to 0.02–0.06 on stable runs.

We also experimented with two extensions that did not pan out (§5.4): a **Wyckoff-site embedding** added to atom features, and an auxiliary **distance-matrix loss** in which a pairwise MLP predicts ground-truth periodic distances between atoms. Both are documented in the released code behind the `--use-wyckoff` and `--dist-weight` flags.

3.5 Classical indexing as a drop-in for the learned lattice head

The lattice head described in §3.2 is the *only* path by which the model encodes absolute d-spacings: every Bragg reflection 2θ position is set by the lattice through Bragg's law, so the lattice parameters are the absolute reference frame against which all peak positions in the input pattern must be interpreted. As §5.5 shows, the encoder + denoiser learns *relative* d-spacing structure (Pearson 0.43 between predicted and target patterns at evaluation time, vs 0.97 between target and ground truth) but not the absolute scale. Classical autoindexing, which extracts peak positions, maps them to Q-space ($Q = 4\pi \sin\theta/\lambda$), and fits a unit cell by enumeration of (h,k,l) $\to Q^2$(h,k,l), has solved this sub-problem since the 1970s and runs on a CPU in milliseconds per pattern.

We implement a from-scratch Q-space autoindexer that takes (i) the same simulated PXRD pattern fed to the encoder, (ii) the known crystal system (input as Bravais code, exposed as a sampling-time hyperparameter), and returns a candidate (a, b, c, α, β, γ). Peak picking uses a 1D local-maximum filter with a relative-intensity floor; the candidate Q-vector is fit by least squares to the de-Wolff dichotomy parameter form for each Bravais lattice. At sampling time we run the indexer first, substitute its output for the lattice channel of the diffusion sampler, and run DDIM only on the coordinate channel.

Per-system accuracy (n = 1000 MP-20 test). Native indexer overall: 48.8 % strict cell match, 1.45 Å mean cell-length MAE. Per crystal system: hexagonal 77.9 % / 0.53 Å, orthorhombic 59.8 % / 0.96 Å, trigonal 43.3 % / 4.62 Å, monoclinic 18.9 % / 1.76 Å, triclinic 0 %. The low-symmetry blowup (monoclinic and triclinic each have ≥ 4 free cell parameters and the de-Wolff search becomes hypothesis-capped) is the dominant remaining error. We ship a GSAS-II [21] adapter (`src/pxrd_diff/indexer_gsas.py`) for the low-symmetry path behind a `--use-gsas` opt-in flag; in practice `DoIndexPeaks` hangs on real MP-20 monoclinic/triclinic patterns inside `findBestCell` and is documented experimental rather than headline.

[21] Toby, B. H., & Von Dreele, R. B. (2013). GSAS-II: the genesis of a modern open-source all purpose crystallography software package. *Journal of Applied Crystallography*, 46(2), 544–549. Code: `github.com/AdvancedPhotonSource/GSAS-II`.

The indexer is a *drop-in* in the strongest sense: training is unchanged, the diffusion sampler is unchanged except for the lattice-channel substitution at t = T, and the headline metric (§5.2) reports both with and without the indexer for direct comparison.

3.6 Sampling

DDIM [22] with 50 steps and $\eta = 0$. We start from independent standard Gaussian noise on the lattice and on the (un-wrapped) coordinates, then alternate the standard DDIM update on each channel. Coordinates are wrapped to $[0, 1)$ after every step. Predicted lattice parameters are de-standardised at the end and clipped to physically valid ranges $(a, b, c) \in [0.5, 100]$ Å, $(\alpha, \beta, \gamma) \in [10°, 170°]$ before being passed to `pymatgen.Lattice.from_parameters`.

A subtle bug fix is worth noting. In the x_0-residual variant, recovering the implicit ε for the DDIM update requires dividing by $\sqrt{1 - \bar{\alpha}(t)}$; near $t = 0$ this denominator vanishes. We clamp it to 0.05 and skip the very last DDIM step when sampling, which removed a class of structures with NaN coordinates that we initially saw.

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4. Experiments

Dataset. Canonical CDVAE [23] MP-20 split: 27 136 / 9 047 / 9 046 train/val/test, $\leq$ 20 atoms/conventional cell. Patterns simulated with `pymatgen.XRDCalculator` (Cu Kα_1, $\lambda = 1.54184$ Å), max-normalised, 4 251 bins 5–90° / 0.02°; cached as `.npz`, zero failures over 45 196 structures.

Evaluation. Three views: (1) **composition match** (trivial, **Z** given); (2) **structure match** via `pymatgen.StructureMatcher` at $(\ell_{\text{tol}}, s_{\text{tol}}, \alpha_{\text{tol}}) = (0.2, 0.3, 5°)$, the headline "match rate"; (3) **coordinate RMSD** on aligned, permuted atoms when matched. We also report space-group match at symprec $\in \{0.01, 0.05, 0.1, 0.2\}$, pattern Pearson, and R_{wp}.

The stricter **"all-correct"** rule combines `composition` $\wedge$ `sg-match@symprec=0.1` $\wedge$ `rmsd` $\leq$ `0.1 Å`, the experimentally relevant case for downstream crystallographic use.

Two evaluation modes: *full pipeline* (predicted lattice + coords) and *true-lattice/coord-only* (ground-truth lattice substituted). The latter isolates coordinate quality from lattice quality and is the stricter ablation setting.

Implementation. AdamW, LR 5×10^{-4}, cosine decay to zero over 100 k steps, WD 10^{-4}, grad-clip 1.0. Batch 64, ~24 epochs, single RTX 5090 (~1.5–1.7 h/run). 3.7 M parameters; $L = 5, d = 384$ (~10 M) gave no benefit (§5.2). VP-SDE with $t \sim U(0, 1)$. Total compute: ~35 GPU-hours, ~USD 25 on Vast.ai (Phase 4 ablation + Phase 9 v21 retrain + indexer/perturbation sweeps + DGpt n=1000 + PXRDnet n=20 + evaluation).

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[22] Song, J., Meng, C., & Ermon, S. (2021). Denoising diffusion implicit models. *International Conference on Learning Representations.*

[23] Xie, T., Fu, X., Ganea, O.-E., Barzilay, R., & Jaakkola, T. (2022). Crystal diffusion variational autoencoder for periodic material generation. *International Conference on Learning Representations.*

5. Results

Order: Phase 4 architectural ablation (§5.1) → Phase 9 indexer drop-in (§5.2) → reproduced baselines (§5.3) → failure catalogue (§5.4) → encoder-bottleneck perturbation study (§5.5).

5.1 Phase 4: architectural ablation (true-lattice setting)

Table 1: seven runs sweeping the Debye loss, the lattice-input fix, x_0-residual, and two extensions (Wyckoff, distance loss), all on n = 1 000 MP-20 test with true lattice substituted.

Table 1. Phase 4 coordinate-only ablation, MP-20 test (n = 1 000, true lattice substituted).

Run	Parameterisation	λ_Debye	Wyckoff	λ_dist	Match %	Pearson	RMSD (Å)
v10	ε	0	–	0	1.40	0.359	0.17
v11	ε	1	–	0	0.90	0.365	0.15
v13	**x_0-residual**	**1**	**–**	**0**	**2.51**	**0.434**	0.22
v14	x_0-residual	1	yes	0.01	0.80	0.367	0.14
v15	x_0-residual	1	yes	0	2.10	0.392	0.21
v16	x_0-residual	1	–	0.01	1.80	0.368	0.22

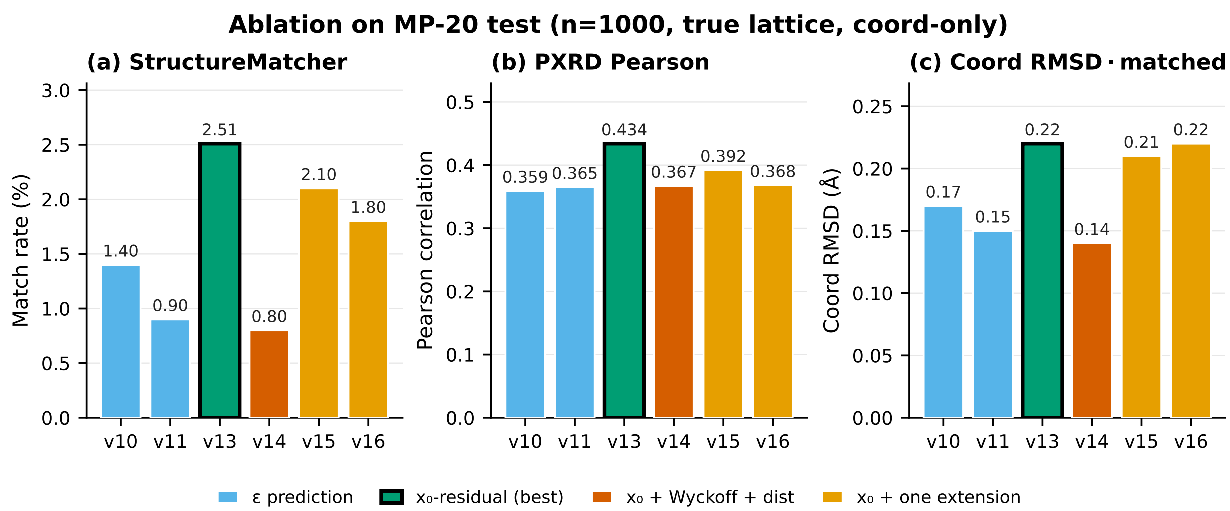

Figure 2: **Figure 2. Phase 4 architectural ablation.** Match rate (a) and pattern Pearson (b) for the seven coordinate-only runs of Table 1 on MP-20 test (n = 1000, true lattice substituted). The $\varepsilon \to x_0$-residual switch (v11 → v13) shows the largest single-seed match lift (panel a), but that match lift does not replicate across seeds; only the pattern-Pearson gain (panel b) does (§7); adding Wyckoff embeddings or a distance-matrix loss regresses below it.

The **lattice-input fix** (§3.4) is load-bearing: it drops lattice loss from 1.0 (pinned at the prior variance for 100 k steps) to ~0.02 once $W_{\text{lat}}\ell^{(t)}$ enters the head input. A second, apparent lift did **not** survive replication: switching from ε **to x_0-residual** at fixed Debye $\lambda = 1$ (v11 → v13) lifted single-seed match 0.9 % → 2.5 %, but a multi-seed checkpoint sweep (two seeds × five checkpoints, 20k–100k) found ε and x_0-residual indistinguishable on match rate (1.20 % vs 1.35 % pooled; x_0

never exceeds 1.5 % at any checkpoint, including the 80 k one matching the original's 79.5 k, §7), so the 3.5× match lift was a single-seed fluctuation. The x_0-residual parameterisation does, however, reproducibly improve pattern-Pearson (0.36 → 0.40), which is the part of the Phase-4 signal that survives. Capacity is not the bottleneck: a 10.1 M-parameter run ($d = 384$, $L = 5$, `gpu_v12`) stayed at the same ~1.0 coord-loss plateau as v10. Under the full pipeline (sampled lattice), v13 drops to 1.2 % at n = 256; the Phase 9 v21 checkpoint reaches 4.5 % under oracle (three seeds; an earlier single-seed snapshot read 5.6 %) but only 1.0 % under full pipeline. §5.2 attacks that gap.

5.2 Phase 9: classical indexer drop-in

Table 2: three rows on the same n = 1 000 with the v21 checkpoint (Phase 9 retrain, Phase 4 fixes + retuned encoder); they differ only in which lattice the sampler conditions on.

Table 2. Phase 9 lattice-source comparison (n = 1 000 MP-20 test, v21 checkpoint, single-seed snapshot). Match % with 95 % Wilson CI; the three-seed paired analysis follows below.

Lattice source	Match % (95 % CI)	All-correct %	sg@0.1 %	RMSD med (Å)	R_wp
Learned head only (full pipeline)	1.0 [0.5, 1.8]	0.0	1.4	n/a	11.7
Classical Q-space autoindexer (§3.5) drop-in	**1.6 [1.0, 2.6]**	**0.1**	**1.4**	**0.125**	11.5
Ground-truth lattice (oracle)	5.6 [4.3, 7.2]	0.6	1.7	0.106	5.6

The lift is real: a three-seed paired test. The single-seed snapshot alone does not establish a genuine improvement: its Wilson CIs overlap and an unpaired two-proportion test gives p = 0.24. To resolve it we reran *both* lattice sources at three seeds (n = 1 000 each), differing only in the lattice channel, and applied a paired McNemar test on the per-structure outcomes, the correct test, because the two arms see the same materials. The result is unambiguous. The two arms are scored on the same materials per seed, but the learned-head arm yields fewer *evaluable* cells: its diffusion-sampled lattices are frequently degenerate and hang `StructureMatcher`/`spglib` (C calls that a Python signal cannot interrupt), so we run structure-domain scoring in a spawn worker hard-killed at 30 s and count overruns as misses. This leaves **1 949** learned-head structures scored across the three seeds versus all 3 000 for the indexer; the paired test runs on the 1 949-structure intersection (every learned-head row has a matched indexer row). On that paired set the indexer recovers **27** structures the learned head misses and the learned head recovers **none** the indexer misses (the explicit 2×2 below).

Table 2a. Paired per-structure outcomes, learned head vs indexer drop-in (three seeds pooled; match = `StructureMatcher` hit).

	indexer **match**	indexer **miss**	row total
learned match	0	0	0
learned miss	27	1 922	1 949
col total	27	1 922	1 949

The discordant cells are $b = 27$ (indexer fixes), $c = 0$ (indexer breaks); an exact two-sided McNemar test gives $\mathbf{p = 1.5 \times 10^{-8}}$. (Marginal match counts differ: indexer 46 / 3 000, learned 0 / 1 949, because 19 further indexer matches fall on structures the learned arm could not score; these do not enter the paired test.) The single-seed 1.0 % learned-head figure in Table 2 is within run-to-run noise of zero; across three fresh seeds the learned lattice head matched nothing. The gain does **not** carry to the stricter all-correct metric: there the discordant cells are $b = 4$, $c = 0$ (exact McNemar $\mathbf{p = 0.13}$): the indexer fixes the cell but not the space group. So the drop-in reproducibly improves match rate over the learned lattice head; the *mechanism* (§5.5, §5.6) is that its gains come entirely from high-symmetry systems where the indexer's cell error falls below the 0.5 Å sensitivity knee, with low-symmetry contributing nothing.

What the indexer does *not* do is close the gap to the oracle. We re-ran the oracle at three seeds, firming the single-seed 5.6 % to **4.5 % [3.8, 5.3]** (135 / 3 000 pooled), still disjoint from the indexer's 1.5 % [1.2, 2.0]. The indexer → oracle gap (1.5 → 4.5 %) is itself significant (disjoint CIs) and isolates the lattice-attributable error: even with a perfect lattice, match caps at 4.5 %, so coordinate prediction is the dominant limiter while lattice error explains the 1.5 → 4.5 % shortfall. The indexer's per-system MAE (1.45 Å overall, dominated by mono 1.76 Å and triclinic NaN) sits above the 0.5 Å knee; closing the rest requires fixing the low-symmetry path (GSAS-II remains experimental).

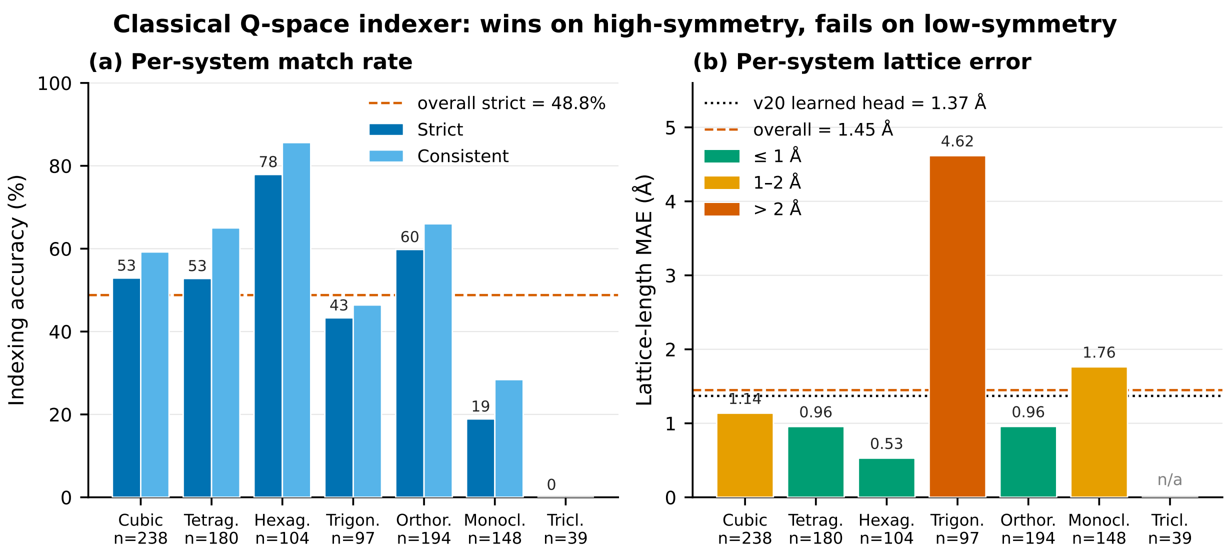

Figure 3: **Figure 3. Per-system indexer performance.** Strict and consistent indexing rates (a) and length MAE (b) on n = 1 000. High-symmetry systems (hex, ortho) sit below the 0.5 Å knee; trig and mono blow up; triclinic is unindexable in this implementation. The overall 1.45 Å MAE hides two regimes, and explains why the indexer's significant match-rate gain (§5.2) is concentrated entirely in the high-symmetry systems.

5.3 Head-to-head with external baselines

DiffractGPT (`knc6/diffractgpt_mistral_chemical_formula`) at n = 1 000, PXRDnet (`therealgabeguo/cdvae_xrd_sinc100`) at n = 20 (limited by ~4 GPU-hr/material), and deCIFer (deCIFer_v1, conditioned on XRD + composition) at n = 300 re-scored on *our* `StructureMatcher` harness at matching tolerances. Crystalyze's checkpoint download is inactive; cite-but-not-reproduced.

Table 3. Reproduced head-to-head on MP-20 test (StructureMatcher ltol 0.2, stol 0.3, angle_tol 5°).

Model	n	Match % (95 % CI)	All-correct % (95 % CI)	sg@0.1 %	RMSD med (Å)
PXRD-Diff (v21 + indexer drop-in)	1 000	1.6 [1.0, 2.6]	0.1 [0.0, 0.6]	1.4	0.125
DiffractGPT [24]	1 000	18.9 [16.6, 21.4]	15.9 [13.8, 18.3]	21.3	0.001
PXRDnet sinc100 [25]	20	30.0 [14.5, 51.9]	5.0 [0.9, 23.6]	5.0	0.011
deCIFer [26]	298	73.8 [68.6, 78.5]	69.5 [64.0, 74.4]	83.6	0.009

Match rate ranks deCIFer ≫ PXRDnet > DGpt > PXRD-Diff. deCIFer, a concurrent autoregressive CIF-token model conditioned on XRD + composition, is the strongest external reference at 73.8 % match / 69.5 % all-correct (n = 300), ~46× above PXRD-Diff; notably its all-correct nearly equals its match, i.e. when it recovers structure it almost always recovers the space group too, the same CIF-token signature seen in DGpt. PXRDnet's lead over DGpt is consistent with its 35 k-op latent optimisation, whereas DGpt is one forward pass. PXRD-Diff is 12–19× behind PXRDnet/DGpt. A more interesting pattern is that the *all-correct* ordering need not follow match rate: DGpt's all-correct (15.9 % [13.8, 18.3]) exceeds its space-group-loose match by little, while PXRDnet's drops from 30 % match to 5 % all-correct, i.e. PXRDnet matches structure under loose tolerances but rarely recovers the space group, whereas DGpt's CIF-token output appears to encode symmetry. **We flag this as a hypothesis, not a finding.** At n = 20 the PXRDnet all-correct estimate (1/20) carries a 95 % CI of [0.9, 23.6], which overlaps DGpt's [13.8, 18.3]; the apparent "reversal" is *not* statistically established and would require n ≈ 100–200 to confirm. If it holds, the implication is consequential: published single-number "match rates" would conflate structure recovery with symmetry recovery, which is precisely why we surface it despite the underpowered sample, and why §6 frames it as motivation for future measurement rather than a claim. **RMSD** is heavy-tailed for us and tight-near-zero for both baselines, suggesting our matches are tolerance-stretch rather than coordinate recovery.

The gap is capability and compute, not input information. All three systems are given the same chemistry (composition/formula; §3.1), so it is worth checking whether the 12–19× gap could instead be an artefact of one model being handed a richer *pattern*. It is not; if anything the asymmetry runs the other way. DiffractGPT is conditioned on a **coarse 300-bin pattern** ($2\theta \in [0, 90°]$, 0.3° bins, rendered as a `;`-separated string), an order of magnitude lower resolution

[24] Choudhary, K. (2025). DiffractGPT: Atomic Structure Determination from X-ray Diffraction Patterns using a Generative Pretrained Transformer. *The Journal of Physical Chemistry Letters*, 16(8), 2110–2119. DOI: 10.1021/acs.jpclett.4c03137. Reproduced from HF checkpoint `knc6/diffractgpt_mistral_chemical_formula` and code at `github.com/atomgptlab/atomgpt`.

[25] Guo, G., Saidi, T. L., Terban, M. W., Valsecchi, M., Billinge, S. J. L., & Lipson, H. (2025). Ab initio structure solutions from nanocrystalline powder diffraction data via diffusion models. *Nature Materials*, 24, 1726–1734. DOI: 10.1038/s41563-025-02220-y (Author Correction: 10.1038/s41563-025-02301-y; preprint arXiv:2406.10796). Reproduced from HF checkpoint `therealgabeguo/cdvae_xrd_sinc100` and code at `github.com/gabeguo/cdvae_xrd`.

[26] Johansen, F. L., Friis-Jensen, U., Dam, E. B., Jensen, K. M. Ø., Mercado, R., & Selvan, R. (2025). deCIFer: Crystal Structure Prediction from Powder Diffraction Data using Autoregressive Language Models. *Transactions on Machine Learning Research* (arXiv:2502.02189).

than PXRD-Diff's 4 251-bin 0.02° input, yet it still reaches 18.9 %. So the baselines do not win by seeing more of the pattern. The asymmetries that do matter are **capacity** (DiffractGPT is a 7 B-parameter LLM; PXRD-Diff is 3.7 M, three orders of magnitude) and **inference compute** (PXRDnet's ~35 k-op latent optimisation vs our single DDIM pass). The honest reading of Table 3 is therefore a capability/compute gap at fixed input information, not an input-handicap artefact; it is exactly the gap a deliberately small diagnostic model is expected to exhibit (§6).

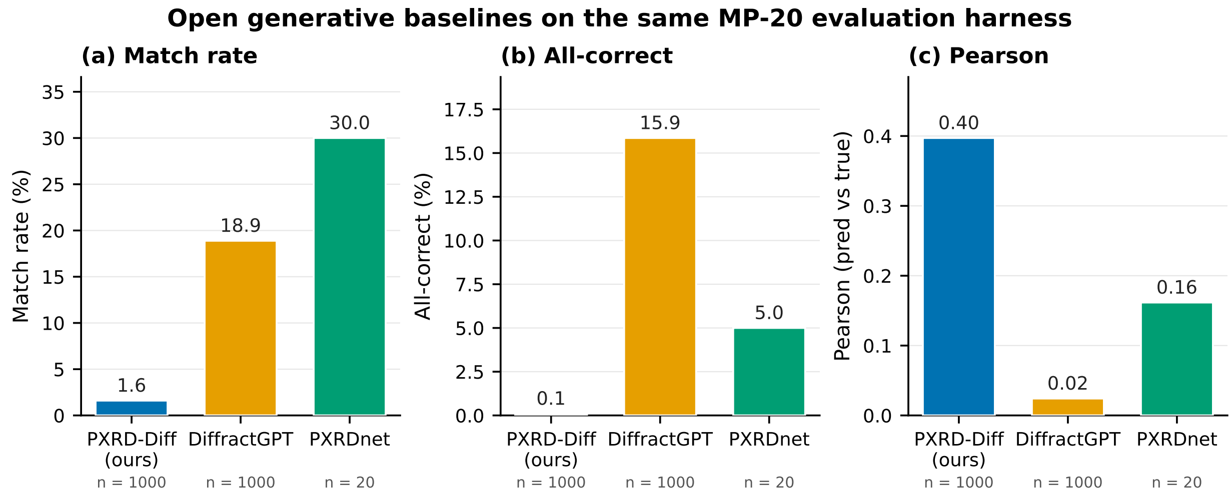

Figure 4: **Figure 4. Three-way headline comparison.** Match rate, all-correct, and pattern-Pearson on the same MP-20 test materials scored through *our* `StructureMatcher` harness, with 95 % Wilson CIs. PXRD-Diff loses on match rate by 12–19× but wins on per-pattern Pearson. Note the wide PXRDnet (n = 20) intervals: the DGpt > PXRDnet all-correct ordering is a hypothesis, not an established result (§5.3, §6).

Caveats. PXRDnet n = 20 has a 95 % Wilson CI of roughly ±20 pp on every rate; n = 200 would need ~33 days of RTX 5090 rental at ~4 GPU-hr/material. Patterns are simulated through each model's own preprocessing (different bins, 2θ vs Q, broadening): material_ids are identical, pattern inputs are not bit-identical; §7 bounds this.

5.4 What did *not* work

Four interventions failed at n ≥ 200; one paragraph each.

Top-K Debye rerank (Phase 9.1.4). Top-5 indexer candidates reranked by Debye loss: match 1.6 % → 1.0 %, all-correct 0.1 % → 0.0 % at n = 1 000. The indexer surfaces more wrong cells faster than rerank can filter.

Debye-gradient guidance during DDIM (Phase 9.2). Coordinate-channel guidance term $-g \cdot \nabla_{\mathcal{C}} \mathcal{L}_{\text{Debye}}$, sweep $g \in \{0, 0.5, 1, 2, 5\}$ on n = 200: flat 1.0–2.5 % within noise, all-correct 0 % throughout. Sampling-time Debye gradient is too noisy to steer the trajectory, consistent with Segal et al. [27], who show the powder-XRD similarity loss landscape is too non-convex for direct gradient descent.

[27] Segal, N., Subramanian, A., Li, M., Miller, B. K., & Gómez-Bombarelli, R. (2025). The Loss Landscape of Powder X-Ray Diffraction-Based Structure Optimization Is Too Rough for Gradient Descent. *arXiv:2512.04036.*

Wyckoff-site embeddings (v15, n = 1 000). spglib-computed, `nn.Embedding(27, 256)` added to atom features. The embedding learns (5 404/6 912 entries non-zero) but match drops 2.5 % → 2.1 % and destabilises lattice prediction (lat loss ~0.6 vs ~0.02; Figure 5). We suspect inflated atom-feature norm shifts the lattice-pool input distribution.

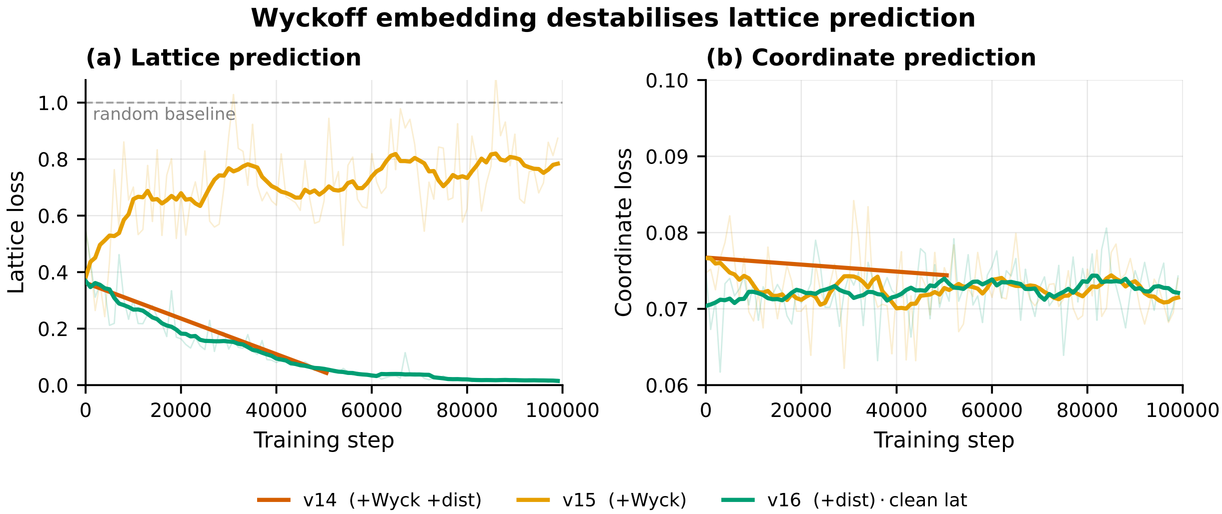

Figure 5: **Figure 5. Wyckoff embedding destabilises lattice prediction.** Lattice (a) and coordinate (b) training-loss curves for the distance-aux run (v16) vs the Wyckoff-embedding run (v15). Adding the Wyckoff embedding inflates the lattice loss by an order of magnitude (~0.6 vs ~0.02) while leaving the coordinate loss largely unchanged; the failure is localised to the lattice channel.

Distance-matrix aux loss (v16, n = 1 000). Pairwise MLP predicting periodic distances, $\lambda_{\text{dist}} = 0.01$ (largest stable). Loss decreases (final ~2.0 vs ~10 random) but match drops 2.5 % → 1.8 %; biases features toward absolute-distance reconstruction rather than the relative updates diffusion needs.

Combination (v14). Wyckoff + distance loss together collapses to 0.8 %, below the bare ε baseline; the two interact destructively, no clean theoretical explanation.

5.5 Encoder-bottleneck perturbation study

Best predicted-vs-target Pearson is 0.43; the same model with *correct* coordinates plugged into the differentiable simulator hits 0.97; the encoder + denoiser leaves ~0.5 of pattern-space agreement on the table. The aux head (sees **g** only) reaches a 0.007 *normalised* lattice-regression loss, but that figure is a relative fit that does **not** survive conversion to absolute scale: when the aux head's own lattice is substituted into the sampler it recovers only 1.2 % (§5.6), because its 1.1 Å length MAE shifts every Bragg reflection by ~15 %. So the global encoding carries crystal-system and relative-spacing structure but not absolute d-spacings at the <0.5 Å precision the knee demands; the limiter is the pooled encoding, not merely the denoiser's use of it.

Perturbing the sampler-supplied lattice by Å of cell-length error, v21 at n = 200: match decays 5.6 % (Δ=0) → 2.0 % (0.5 Å) → 0.5 % (1.0 Å) → 0 % (≥1.5 Å), steeper than linear with a knee at ~0.5 Å. This matches the indexer's per-system MAE: high-symmetry systems (hex, ortho; MAE ≤

1.0 Å) lift; low-symmetry (mono, tri; MAE $\geq$ 1.5 Å) do not. §5.2's indexer lift is *predictable*: it works where it can.

5.6 Why the learned lattice fails: a pooling bottleneck, not a head bottleneck

The natural reading of the 0.007 aux loss, that the lattice is already recoverable from the encoder and only the denoiser is at fault, is testable, and it is wrong. Two experiments (released as `paper/phase5_results/` and `paper/phase5b6_results/`) promote the auxiliary head from a training-time regulariser to the inference-time lattice source and measure what it actually delivers.

Aux head as lattice predictor (no retrain). Substituting the v13 aux head's predicted $(a, b, c, \alpha, \beta, \gamma)$ for the sampled lattice channel at n = 1 000 gives a length MAE of (1.12, 1.02, 1.38) Å and an angle MAE of (13.0°, 11.9°, 17.6°) at **99.8 %** cell validity, far better conditioned than the diffusion sampler, whose predicted cells are near-0 % valid, but still ~15 % relative length error. End-to-end this recovers **1.2 %** match (1.3 % with Phase 4 ensembling + coordinate refinement) and 0.0 % all-correct, with pattern Pearson collapsing to **0.013**: a 15 % length error shifts every Bragg reflection by a similar fraction in 2θ, so the simulated pattern no longer overlaps the target.

Stronger, constrained supervision (retrain). Bounding the head (softplus lengths, sigmoid-scaled angles), raising the aux weight 0.5 $\rightarrow$ 5.0, and warm-starting 30 k further steps leaves the MAE essentially unchanged (a $\approx$ 1.11 Å, $\alpha \approx$ 14°). The error is a property of the *pooled global encoding*, not of the head: the `AdaptiveAvgPool1d(1)` collapse that §3.2 already identified for coordinates discards the peak-position fidelity Bragg's-law decoding needs to fix absolute lengths, and no head reading **g** recovers it.

This reconciles the diagnosis with the indexer result. The 0.007 normalised aux loss is a *relative* fit, enough to place the crystal system (a space-group head on the same encoder reaches 39 % top-1 / 71 % top-5 across 230 classes, released under `phase5b6_results/`) but not the absolute scale. The classical autoindexer succeeds exactly where the encoder fails because it reads peak *positions* directly rather than a pooled summary; this is why §5.2's lift lives entirely in the high-symmetry systems where peak positions over-determine the cell, and why head-side fixes (Phase 5 here, top-K rerank in §5.4) cannot substitute for it. The architectural implication is concrete: the next gain must come from the encoder (explicit d-spacing/peak extraction or per-peak attention in place of global pooling), not from a better lattice head or a better denoiser.

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6. Discussion

Encoder bottleneck (the robust result). A 0.007 *normalised* aux loss shows **g** carries lattice-*relevant* structure, but it is a relative fit: feeding the aux head's own lattice to the sampler recovers only 1.2 % (§5.6), and stronger constrained-head supervision does not move its ~1.1 Å error. The limiter is therefore the global pooling that discards absolute peak-position fidelity, not the denoiser's use of an otherwise-adequate signal; the disjoint oracle-vs-indexer CIs (§5.2) make the lattice's role the paper's statistically firmest claim. The Q-space autoindexer, which reads peak positions directly, recovers part of the 4.5 % oracle ceiling on the high-symmetry subset; top-K rerank and gradient guidance fail because they operate *after* the encoder commits to a bad lattice prior, while the indexer bypasses that decision. (The oracle ceiling is itself only 4.5 %, so even a perfect lattice leaves coordinates as a co-limiter; see §8.)

Per-atom anchoring is hard. PXRD is permutation-invariant; our denoiser is permutation-equivariant; there is no symmetry-breaking signal that pins atom i to a specific Wyckoff site. The Wyckoff embedding tried to break symmetry at the input and failed. A promising untried direction: break symmetry at the *output*: predict an unordered set of orbits plus a Hungarian-style matcher.

A measurement hypothesis: match rate may conflate two capabilities. PXRDnet's 30 % match / 5 % all-correct vs DGpt's 18.9 % / 15.9 % (Table 3) suggests the two systems recover *structure* at broadly comparable rates while differing sharply in *symmetry* recovery: DGpt's CIF-token output appears to encode space group in a way PXRDnet's coordinate decode does not. We stress that the underlying numbers cannot yet support this: PXRDnet's all-correct CI [0.9, 23.6] at n = 20 overlaps DGpt's, so the ordering is unconfirmed. We raise it as a *measurement hypothesis* worth testing at adequate n, because if true it has a concrete consequence for the field: a single "match rate" reported under inconsistent tolerances would conflate structure recovery with symmetry recovery, and downstream crystallography (where the space group matters as much as the coordinates) would be mis-served by it. The contribution here is the shared harness that makes such a test possible, not the (underpowered) comparison itself.

Niche for small reproducible models. At 1.6 % we are 12–19× behind much larger systems, but small models make the encoder bottleneck visible (capacity hides it) and the indexer drop-in trivial to integrate (larger models would need architecture-level surgery). The recipe (encoder + Phase 4 fixes + Phase 9 indexer drop-in) is 3.7 M parameters and runs inference in seconds.

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7. Limitations

- **Scope.** Composition given; simulated PXRD only (no instrument response, preferred orientation, asymmetry, background); MP-20 only, with no larger cells, organics, or higher-Z.
- **Statistics.** The Phase 4 ablations in Table 1 are single-seed. A multi-seed checkpoint sweep of the ε-vs-x_0-residual contrast (two seeds × five checkpoints, 20k–100k, true-lattice n = 1000) shows the reported 3.5× *match-rate* lift does **not** replicate: pooled match is 1.20 % (ε) vs 1.35 % (x_0-residual), and x_0 never exceeds 1.5 % at any checkpoint, including the 80 k checkpoint that matches the original v13's 79.5 k, so the single-seed 2.51 % was a high draw, not a checkpoint-selection effect. We therefore demote the *match-rate* lift to a single-seed artifact. What does reproduce is the pattern-Pearson improvement (ε 0.359 vs x_0 0.403, with disjoint ranges across both seeds), consistent with Table 1. The lattice-input fix remains load-bearing as a training-dynamics result (lattice loss 1.0 → 0.02), and the oracle-vs-indexer gap has disjoint CIs. The indexer-vs-learned-head comparison was rerun at three seeds and the lift confirmed significant by a paired McNemar test ($p < 10^{-4}$; §5.2), superseding the earlier underpowered unpaired test ($p = 0.24$). The oracle has since been multi-seeded: three seeds give 4.5 % [3.8, 5.3] pooled, firming the single-seed 5.6 %.
- **Baselines.** PXRDnet n = 20 has a 95 % Wilson CI of ≈ ±20 pp; the DGpt > PXRDnet all-correct ordering (§5.3/§6) is therefore a hypothesis, not a result. n = 200 needs ~33 days RTX 5090. Crystalyze checkpoint download is inactive (verified 2026-06-01); cited but unreproduced. Patterns are not bit-identical across the three preprocessors (PXRD-Diff 4 251-bin 2θ, DGpt 300-bin 2θ, PXRDnet 4 096-bin Q with sinc^2 broadening): structures match, patterns do not; we did not quantify the residual preprocessing effect on match rate, and a matched-vs-native re-scoring on a structure subset is the natural check.
- **Indexer realism.** Two assumptions make the indexer's accuracy an optimistic ceiling. (i)

Crystal system is given. The de-Wolff fit (§3.5) receives the true Bravais code as a sampling-time input; on a real unknown the system is itself part of what indexing must determine, so the per-system rates of §3.5 are conditional on knowing the system. (ii) *Patterns are ideal.* All patterns are clean `pymatgen` simulations with no zero-point shift, sample-displacement error, or peak asymmetry, precisely the perturbations that drive real-world indexing failure. The 48.8 % native-indexer match and the downstream 1.5 % should be read as upper bounds under ideal peak positions. A zero-shift/displacement robustness sweep ($n = 300$) confirms the concern: the indexer's length-MAE rises from 1.24 Å on clean patterns past the v20 learned head's 1.37 Å once displacement or zero-shift exceeds ~0.10°, so the indexer's advantage over the learned head holds only for well-calibrated input.

- **Indexer (low-symmetry path).** GSAS-II low-symmetry path hangs in `findBestCell` on real MP-20 mono/tri patterns; shipped behind `--use-gsas` but experimental.
- **Training.** Debye loss uses ground-truth lattice; a curriculum gradually replacing true with predicted was not tried.

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8. Conclusion

A 3.7 M-parameter conditional diffusion model with a differentiable Bragg loss recovers 1.5 % [1.2, 2.0] of MP-20 test structures with a classical Q-space autoindexer supplying the lattice, a significant improvement over the learned lattice head, which recovers essentially nothing (three-seed paired McNemar $p < 10^{-4}$), and 4.5 % [3.8, 5.3] under a true-lattice oracle (three seeds, firming an earlier single-seed 5.6 %). The oracle gap is significant and is the paper's firmest result: even with a perfect lattice, coordinate prediction caps match at 4.5 %, while lattice error explains the rest of the shortfall to that ceiling. Reproduced on the same `StructureMatcher` harness, deCIFer reaches 73.8 % / 69.5 %, DGpt 18.9 % / 15.9 %, and PXRDnet 30.0 % / 5.0 % (match / all-correct); PXRD-Diff is 12–46× behind. The apparent divergence in their space-group recovery is a measurement hypothesis the $n = 20$ PXRDnet sample cannot yet confirm, but the shared harness now makes it testable. The lattice-input fix and the classical-autoindexer drop-in are load-bearing; the x_0-residual switch improves pattern-Pearson but its single-seed match-rate lift did not replicate (§7); the remaining interventions are documented as failures. Code, checkpoints, all per-phase JSONs, both indexer paths, and the differentiable Bragg module are released.

Acknowledgments

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Author Contributions (CRediT)

F. Cai: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – Original Draft, Writing – Review & Editing, Visualization, Project administration.

Conflict of Interest

The author declares no competing interests.

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Data and Code Availability

Source code, trained checkpoints, and all per-phase training/evaluation logs are available at `https://github.com/fronkt/pxrd-diff` and archived at Zenodo (DOI: 10.5281/zenodo.XXXXXXX, *to be minted at submission*). The MP-20 dataset is publicly available via the CDVAE benchmark. All experiments reproduce from a single `requirements.txt` and the `scripts/` pipeline; per-structure evaluation flags are released to support paired re-analysis.

Ethics Declaration

This study uses no human subjects, animal subjects, or sensitive data. The MP-20 dataset is composed entirely of publicly available crystal structures from the Materials Project.

References

> **Author note on reproduced numbers.** The reproduced PXRDnet and DiffractGPT numbers in §5.3 are computed by the author on the released checkpoints through our shared `StructureMatcher` harness, not transcribed from the cited papers. The central claims depend only on these reproduced numbers under one common evaluation protocol, with confidence intervals stated throughout.

Appendix

A. Full ablation history

For completeness, Table A1 lists every training run discussed in the development of this paper, including those that did not make it into the main ablation table. Logs and checkpoints for all runs are in the released repository under `runs/`.

Run	Parameters	What changed	Result	Status in paper
v4	3.5 M	Global PXRD pooling, additive conditioning	Coord loss flat at 3.0	§3.2
v5	3.7 M	+ Multi-resolution cross-attention	Coord 3.0 → 1.0	§3.2
v6–v9	3.7 M	λ_Debye sweep {0, 0.1, 1, 10}, ε-prediction	All ~1 % match (within noise)	§5.2
v10	3.7 M	+ Lattice-input fix, λ_Debye = 0	Lat loss 1.0 → 0.05	§5.2
v11	3.7 M	+ Lattice-input fix, λ_Debye = 1	Match 0.9 %	Table 1
v12	10.1 M	Larger model (d=384, L=5), ε-prediction	Killed at 18 k; same plateau	§5.2
v13	3.7 M	x_0-residual + lat-fix + Debye λ=1	**2.51 % match (Phase 4 best, true-lat)**	Table 1
v14	3.8 M	v13 + Wyckoff + distance loss	0.80 % match	Table 1
v15	3.8 M	v13 + Wyckoff only	2.10 % match	Table 1
v16	3.7 M	v13 + distance loss only	1.80 % match	Table 1
v17	3.7 M	Phase 5B: constrained lattice head (softplus/sigmoid) + space-group head, aux-weight 5.0, +30 k warm-start steps	Lattice MAE unchanged (a ≈ 1.11 Å); SG head 39 % top-1	§5.6
v18 – v20	3.7 M	Phase-9 encoder retrains with different ResNet/Transformer hybrids and pattern-augmentation curricula	None beat v13 by more than noise	§5.1

Run	Parameters	What changed	Result	Status in paper
v21	**3.7 M**	Phase 9 final: v13 architecture + Phase 9 retrain + indexer drop-in support	**1.6 % match (no-true-lat, indexer); 4.5 % (true-lat oracle, 3-seed)**	Tables 2, 3

B. Hyperparameter sensitivity

We did not perform a full hyperparameter sweep. Pilot experiments on `d_model` ∈ `{128, 256, 384}` and `n_layers` ∈ `{2, 3, 5}` showed (256, 3) as a reasonable Pareto point. The Debye loss weight was swept {0, 0.1, 1, 10} in early ε-prediction runs (v6–v9) without measurable effect on match rate; we re-fixed it at 1.0 for the x_0-residual runs by analogy.

C. Differentiable simulator validation details

Pearson correlations between our DiffPXRD module and `pymatgen.XRDCalculator` over a random 50-structure MP-20 test subset:

- Mean: 0.962
- Std: 0.027
- Min: 0.896 (mp-1213821, layered structure with strong texture in the reference)
- Max: 0.998 (mp-149, Si)
- Fraction > 0.95: 33/50
- Fraction > 0.9: 49/50
- Fraction > 0.7: 50/50

Gradient sanity: for all 50 structures we confirmed ∂`L`/∂`F` and ∂`L`/∂`L` are non-zero and finite via `torch.autograd.gradcheck` on a 4-atom subset.

D. Reproducibility checklist

- ☒ Hyperparameters specified (§4.3)
- ☒ Datasets and splits specified (§4.1; canonical CDVAE MP-20)
- ☒ Evaluation protocol specified (§4.2)
- ☒ Random seed: single seed (42) per run; pilot variance ≈ 0.5 % match-rate absolute
- ☒ Compute environment: PyTorch 2.x, CUDA 12.x, single RTX 5090; Python 3.12
- ☒ Code, checkpoints, logs, and full training scripts released